

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

Math 4441: Probability and Statistics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Suppose that when a machine is adjusted properly, 50% of the items produced by it are of high quality and the other 50% percent are of medium quality. Suppose, however, that the machine is improperly adjusted during 10% of the time and that, under these conditions, 25% percent of the items produced by it are of high quality and 75% percent are of medium quality.
 - a) Suppose that one item produced by the machine at a certain time is selected at random and inspected. If the item is of high quality, find the probability that the machine was adjusted properly at that time. 10
(CO1)
(PO1)
 - b)
 - i. Suppose that five items produced by the machine at a different time are selected at random and inspected. If four of these items are of high quality and one item is of medium quality, find the probability that the machine was adjusted properly at that time. 10
(CO1)
(PO1)
 - ii. Suppose further one additional item, which was produced by the machine at the same time as the other five items, is selected and found to be of medium quality. Find the new posterior probability that the machine was adjusted properly. 10
(CO1)
(PO1)
2.
 - a) The demand for a certain weekly magazine at a newsstand is a random variable, X , with Probability Mass Function $P_X(x) = (10x)/18, x = 4, 5, 6, 7$. Each of the magazine sells for a price $\$a$ and costs $\$2a/3$ to the owner and the unsold magazines cannot be returned. Find how many magazines should be ordered every week so that the profit will be maximized in the long run. 10
(CO1)
(PO1)
 - b) Let X be a random variable with Probability Density Function:

$$f_X(x) = \begin{cases} \frac{1}{2}e^{-|x|}, & \text{for } -\infty < x < +\infty; \\ 0, & \text{otherwise.} \end{cases}$$
10
(CO1)
(PO1)
 Find the variance and Cumulative Distribution Function of X .
3.
 - a) From the set $\{x : 0 \leq x \leq 1\}$, numbers are selected at random and independently and rounded to three decimal places. Find the probability that
 - i. The number 0.345 is obtained for the first time on the 1000th selection. 7
(CO2)
(PO2)
 - ii. The number 0.345 is obtained for the third time on the 3000th selection. 6
(CO2)
(PO2)
 - b) A box contains 8 red balls and 7 blue balls. You conduct an experiment in which 4 balls are drawn at random, without replacement, from the box. Suppose you repeat this experiment 4 times and each time, you count the number of red and blue balls drawn. Let the random variable X represent the number of times you draw exactly 2 red balls and 2 blue balls. Find the Probability Mass Function (PMF) of X and compute its expected value $E[X]$. 12
(CO2)
(PO2)

Appendix - A

PMF/PDF, expected values and variance of known Random Variables

Families of Distribution	PMF or PDF	Expectation	Variance
Bernoulli $X \sim Ber(p)$	$P_X(x) = \begin{cases} p^x(1-p)^{1-x}, & x = 0, 1 \\ 0, & \text{otherwise.} \end{cases}$	p	$p(1-p)$
Geometric $X \sim Geom(p)$	$P_X(x) = \begin{cases} p(1-p)^{x-1}, & x = 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$
Binomial $X \sim Bin(n, p)$	$P_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, \dots, n \\ 0, & \text{otherwise.} \end{cases}$	np	$np(1-p)$
Pascal/-ve Binomial $X \sim pascal(k, p)$	$P_X(x) = \begin{cases} \binom{x-1}{k-1} p^k (1-p)^{x-k}, & x = k, k+1, \dots \\ 0, & \text{otherwise.} \end{cases}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$
Poisson $X \sim Poisson(\lambda)$	$P_X(x) = \begin{cases} \frac{(\lambda T)^x e^{-\lambda T}}{x!}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$	λT	λT
Hyper Geometric $X \sim HGem(r, g, n)$	$P_X(x) = \begin{cases} \frac{\binom{r}{x} \binom{g}{n-x}}{\binom{r+g}{n}}, & x = 0, 1, \dots, \min(r, n) \\ 0, & \text{otherwise.} \end{cases}$	$\frac{rn}{r+g}$	$\frac{nrg}{(r+g)^2} \left(1 - \frac{n-1}{r+g-1}\right)$
Uniform (discrete) $X \sim unif(a, b)$	$P_X(x) = \begin{cases} \frac{1}{b-a+1}, & x = a, a+1, \dots, b \\ 0, & \text{otherwise.} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)(b-a+1)}{12}$
Exponential $X \sim exp(\lambda)$	$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gaussian $X \sim N(\mu, \sigma^2)$	$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, & -\infty < x < +\infty \\ 0, & \text{otherwise.} \end{cases}$	μ	σ^2
Uniform (Continuous) $X \sim unif(a, b)$	$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise.} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Gamma $X \sim gam(r, \lambda)$	$f_X(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$
Multinomial	$P_{X_1, \dots, X_r}(x_1, \dots, x_r) = \begin{cases} \binom{n}{x_1, \dots, x_r} p_1^{x_1} \dots p_r^{x_r}, & x_i = 0, \dots, n, 0 \leq i \leq r, \sum_{i=1}^r x_i = n \\ 0, & \text{otherwise.} \end{cases}$		
Multivariate Hyper geometric	$P_X(x) = \begin{cases} \frac{\binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_r}{x_r}}{\binom{n_1 + n_2 + \dots + n_r}{x_1 + x_2 + \dots + x_r}}, & x = 0, 1, \dots, \min(r, n) \\ 0, & \text{otherwise.} \end{cases}$		

Appendix - B

General Formulas

Name	Formula
Conditional Probability	$P[A B] = \frac{P[AB]}{P[B]}, P[B] > 0$
Product Rule	$\frac{P[AB]}{P[A_1 \dots A_n]} = \frac{P[A]P[B A]}{P[A_1]P[A_2 A_1] \dots P[A_n A_1 \dots A_{n-1}]}$
Conditional Product Rule	$P[AB C] = P[A C]P[B AC]$
Sum Rule	$P[A] = \sum_{i=1}^n P[A B_i]P[B_i]$
Conditional Sum Rule	$P[A C] = \sum_{i=1}^n P[A B_i, C]P[B_i C]$
Bayes' Theorem	$P[B_i A] = \frac{P[A B_i]P[B_i]}{\sum_{i=1}^n P[A B_i]P[B_i]}$
Conditional Bayes' Theorem	$P[B_i A, C] = \frac{P[A B_i, C]P[B_i C]}{\sum_{i=1}^n P[A B_i, C]P[B_i C]}$
Independence of Events	$P[AB] = P[A]P[B]$
Expected Value	$E[X] = \sum_{x \in S_X} xP_X(x)$ $E[X] = \int_{-\infty}^{+\infty} xf_X(x)dx$
Variance	$Var[X] = E[(X - \mu_X)^2] = E[X^2] - (\mu_X)^2$
Marginal PMF	$P_X(x) = \sum_{y \in S_Y} P_{XY}(x, y)$ $f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x, y)dy$
Covariance	$Cov[X, Y] = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$
Correlation	$r_{XY} = \frac{E[XY]}{Cov[X, Y]}$
Correlation Coefficient	$\rho_{XY} = \frac{Cov[X, Y]}{\sqrt{Var[X]Var[Y]}}$